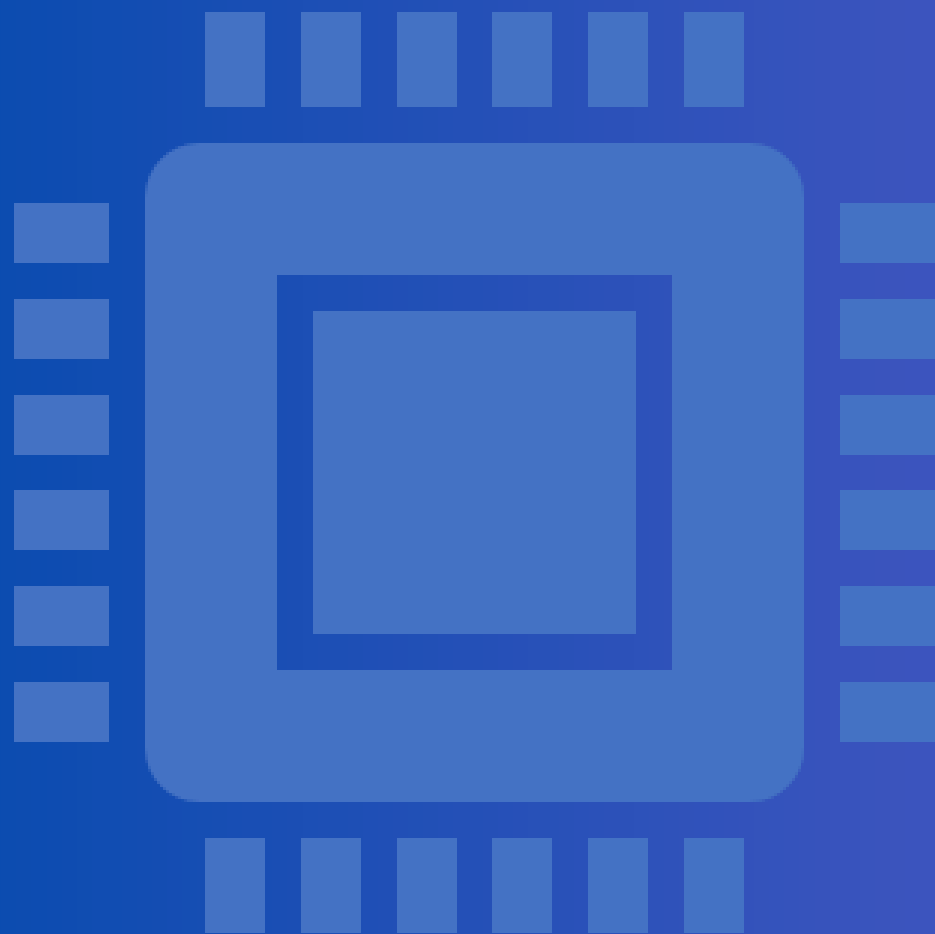


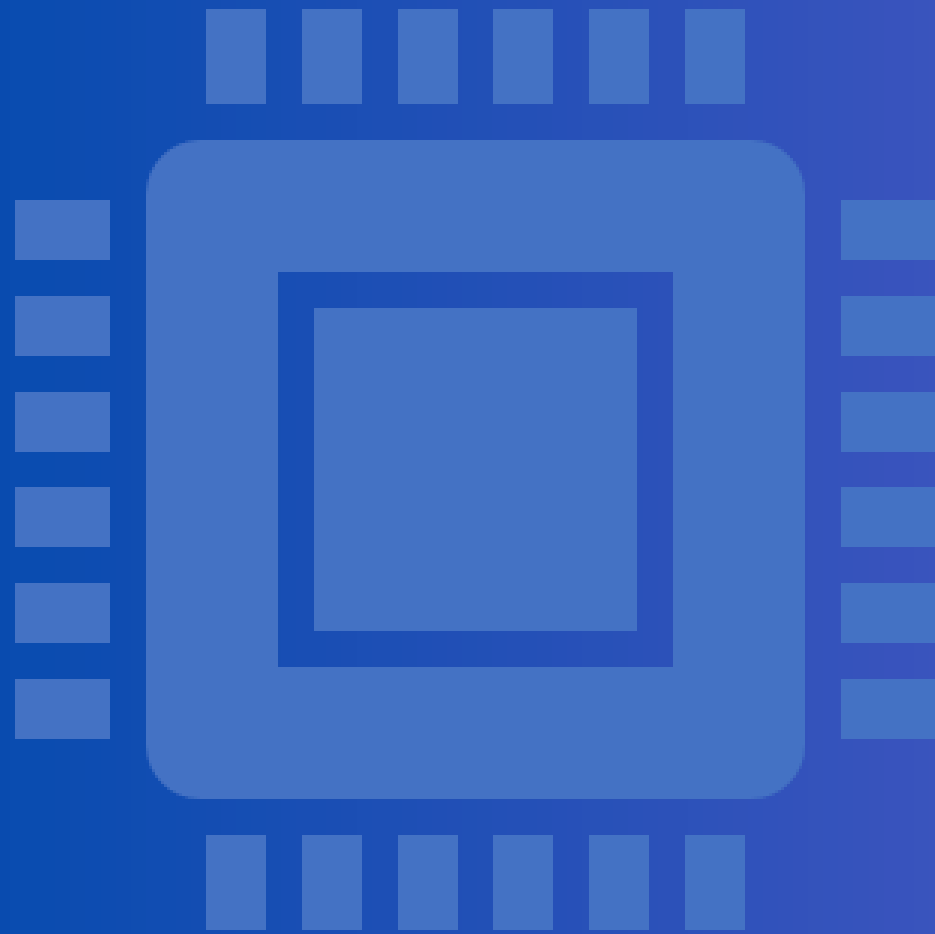
DATA REPRESENTATION CONTINUATION

BITS AND BYTES



- TERMINOLOGY RELATED TO BITS AND BYTES IS EXTENSIVELY USED TO DESCRIBE STORAGE CAPACITY AND NETWORK ACCESS SPEED
- WHAT IS THE DIFFERENCE BETWEEN A BIT AND A BYTE?

BITS AND BYTES



- **TERMINOLOGY RELATED TO BITS AND BYTES IS EXTENSIVELY USED TO DESCRIBE STORAGE CAPACITY AND NETWORK ACCESS SPEED**
- **BIT - BINARY DIGIT**
- **BYTE - GROUP OF 8 BITS , ABBREVIATED AS AN UPPERCASE B**

UNIT TERMINOLOGY CHART

Unit	Abbreviation	Approximate Size
Bit	b	Binary digit, single 1 or 0
Nibble	–	4 bits
Byte	B	8 bits
Kilobyte	KB	1,024 bytes or 10^3
Megabyte	MB	1,024 KB or 10^6
Gigabyte	GB	1,024 MB or 10^9
Terabyte	TB	1,024 GB or 10^{12}
Petabyte	PB	1,024 TB or 10^{15}
Exabyte	EB	1,024 PB or 10^{18}
Zettabyte	ZB	1,024 EB or or 10^{21}

NUMERIC REPRESENTATION

REPRESENTING NUMERIC VALUES

- DECIMAL
 - BINARY
 - HEXADECIMAL
- WHOLE
NUMBERS

INTEGERS,
FLOATING-
POINT

- MODIFY BINARY

REPRESENTING NEGATIVE VALUES

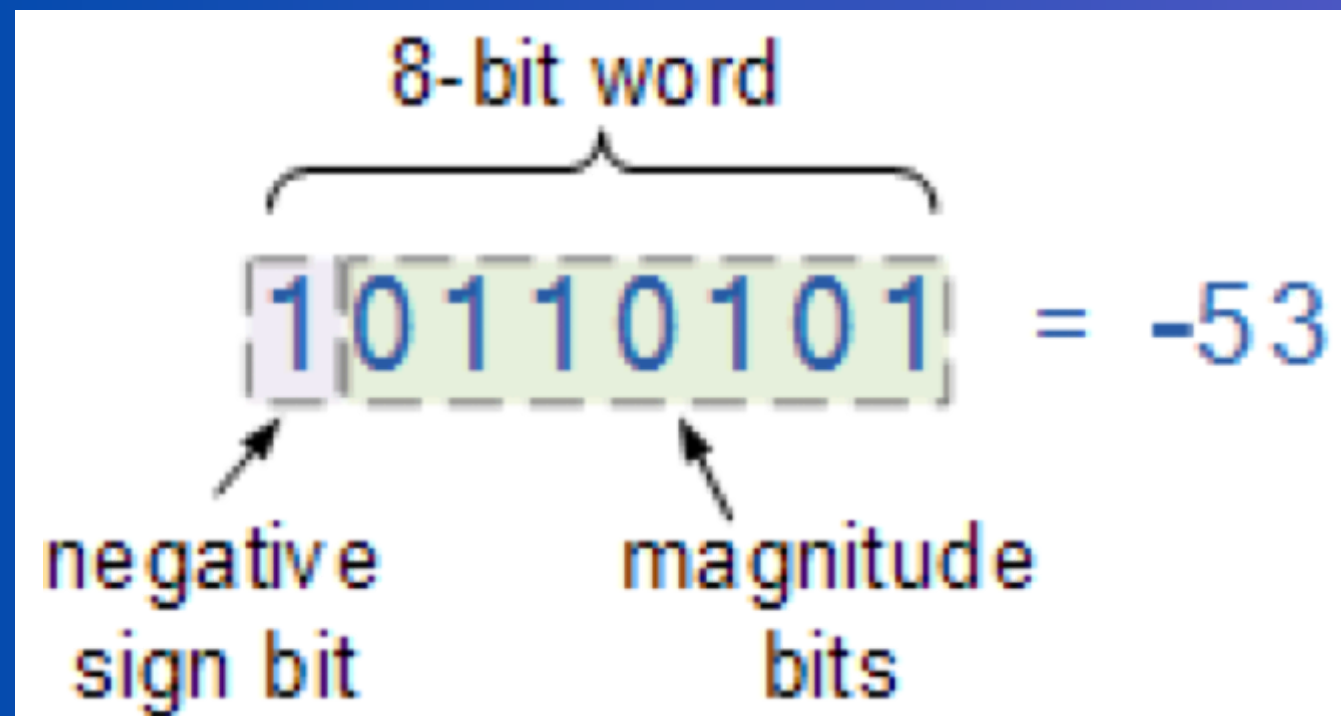
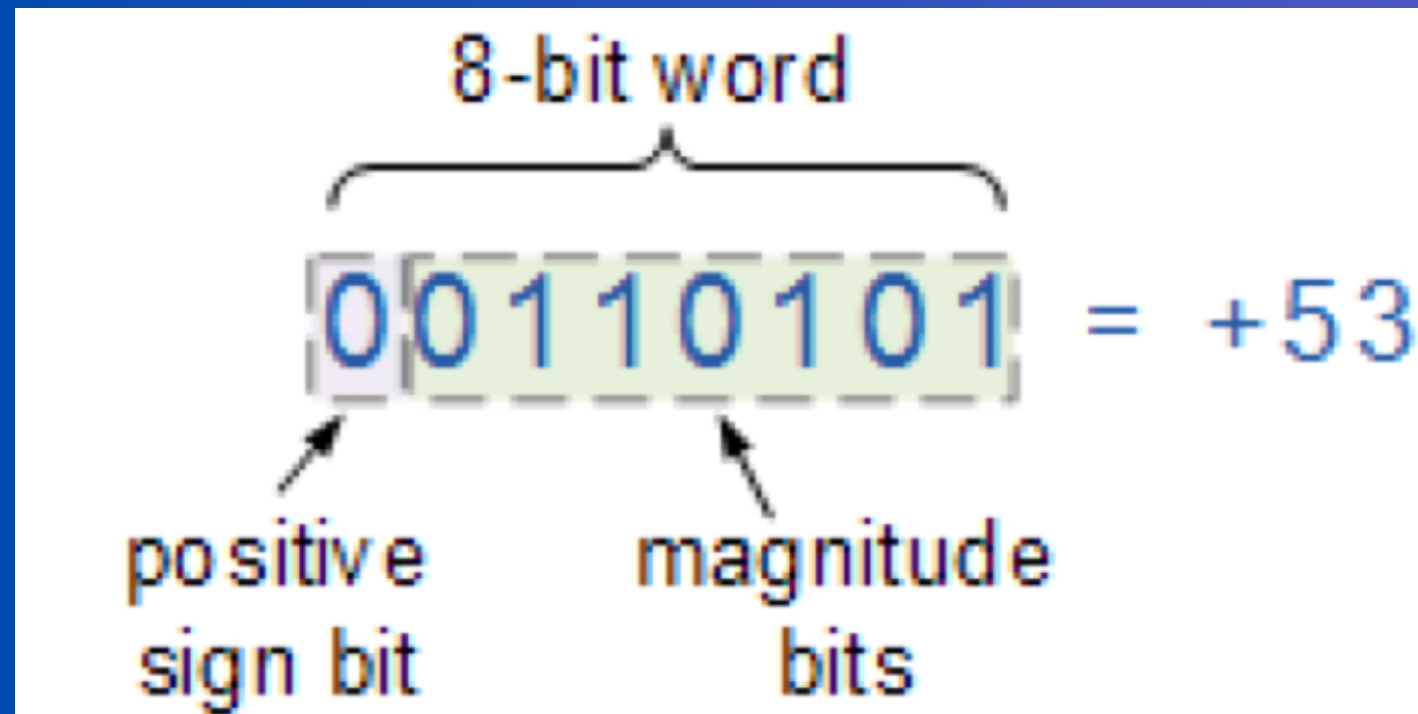
SIGNED-MAGNITUDE(SN)
NOTATION

1's COMPLEMENT

2's COMPLEMENT

SIGNED-MAGNITUDE(SN)

NOTATION

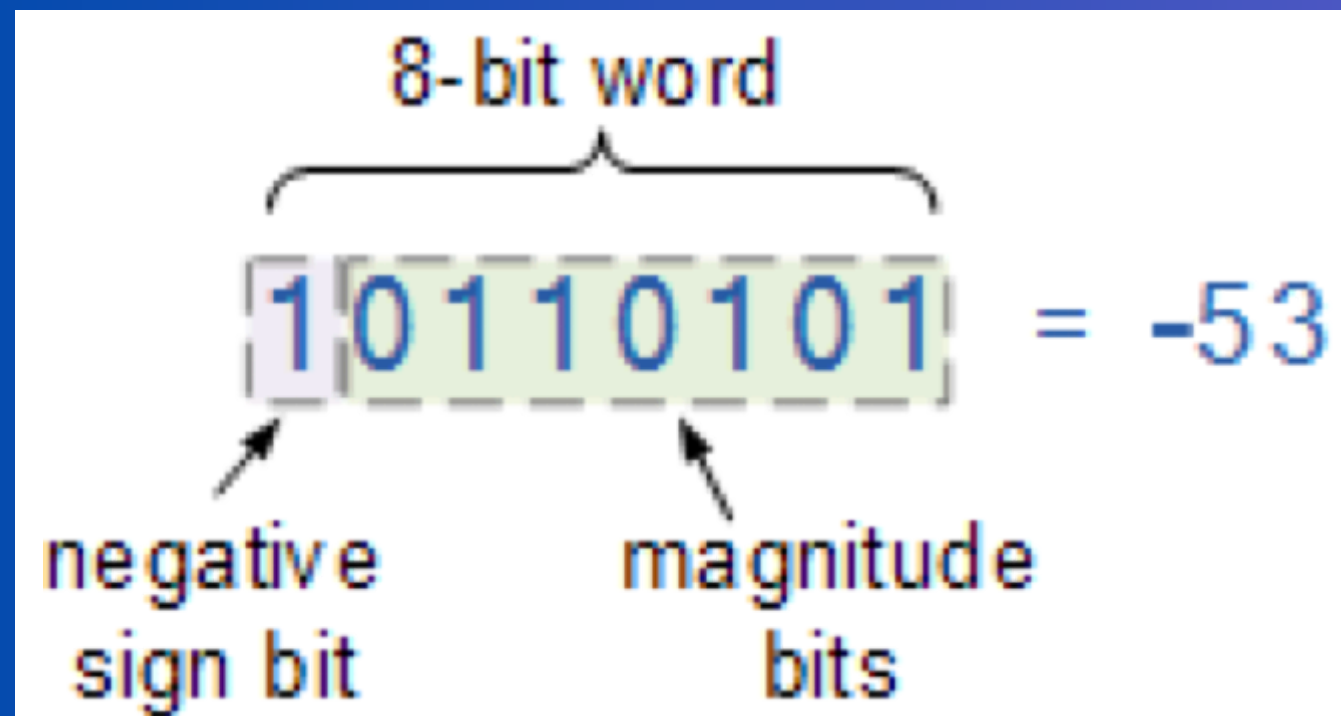
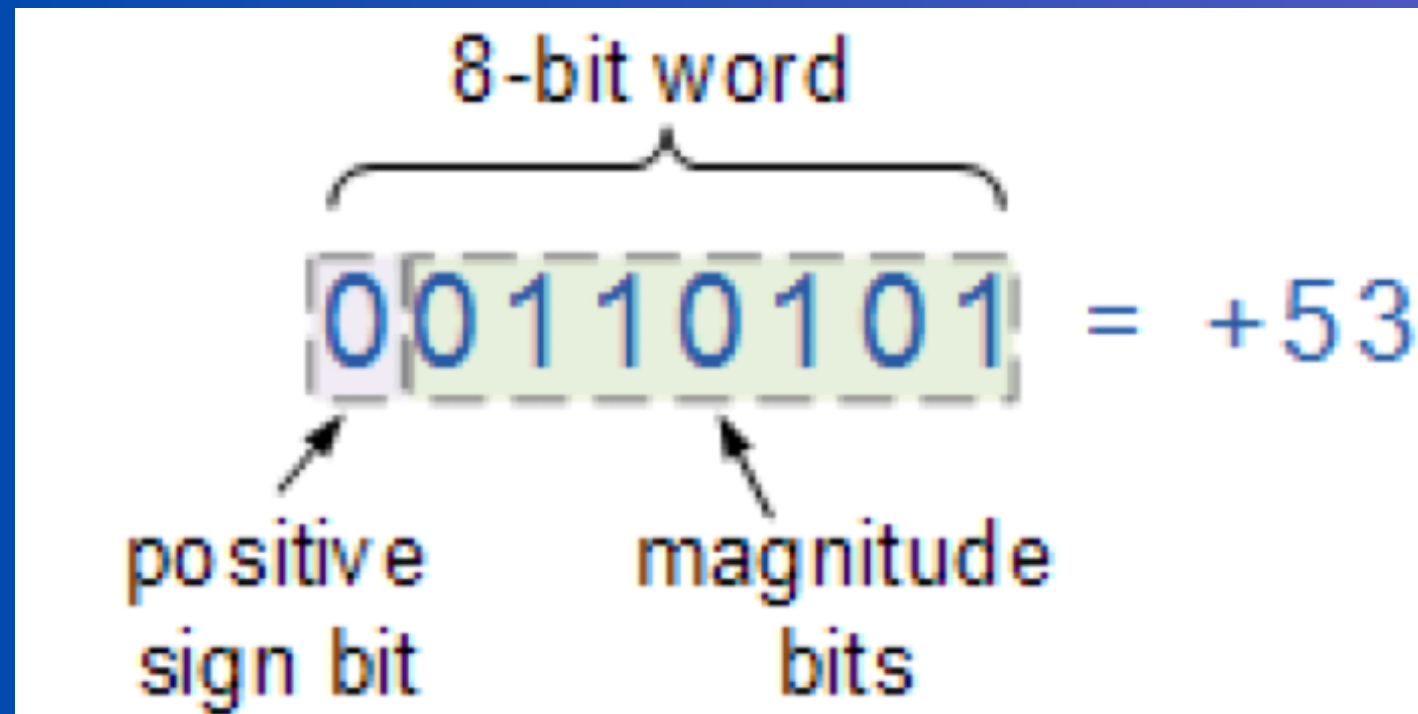


A METHOD OF REPRESENTING SIGNED BINARY NUMBERS WHERE THE LEFTMOST BIT (MSB) INDICATES THE SIGN (0 FOR POSITIVE, 1 FOR NEGATIVE), AND THE REMAINING BITS REPRESENT THE MAGNITUDE (ABSOLUTE VALUE) OF THE NUMBER.

WHAT IS THE PROBLEM WITH SM NOTATION?

SIGNED-MAGNITUDE(SN)

NOTATION



A METHOD OF REPRESENTING SIGNED BINARY NUMBERS WHERE THE LEFTMOST BIT (MSB) INDICATES THE SIGN (0 FOR POSITIVE, 1 FOR NEGATIVE), AND THE REMAINING BITS REPRESENT THE MAGNITUDE (ABSOLUTE VALUE) OF THE NUMBER.

THERE ARE TWO REPRESENTATIONS FOR 0, A POSITIVE AND A NEGATIVE REPRESENTATION

1'S COMPLEMENT

A BINARY REPRESENTATION OF NEGATIVE NUMBERS
WHERE ALL BITS ARE INVERTED ($0 \rightarrow 1, 1 \rightarrow 0$).

- POSITIVE NUMBERS REMAIN UNCHANGED
- POSITIVE NUMBER COMPLEMENTS ALWAYS START WITH 1

EXAMPLE: 8-BIT REPRESENTATION OF 12

+12: 0000 1100

-12: 1111 0011

WHAT IS THE PROBLEM WITH
1'S COMPLEMENT?

1's COMPLEMENT

A BINARY REPRESENTATION OF NEGATIVE NUMBERS
WHERE ALL BITS ARE INVERTED ($0 \rightarrow 1, 1 \rightarrow 0$).

- POSITIVE NUMBERS REMAIN UNCHANGED
- POSITIVE NUMBER COMPLEMENTS ALWAYS START WITH 1

EXAMPLE: 8-BIT REPRESENTATION OF 12

+12: 0000 1100

-12: 1111 0011

IT ALSO HAS TWO
REPRESENTATIONS FOR 0.

+0: 0000

- 0: 1111

2's COMPLEMENT

A BINARY REPRESENTATION OF NEGATIVE NUMBERS OBTAINED BY INVERTING ALL BITS (1's COMPLEMENT) AND ADDING 1 OR BEGINNING INVERSION AFTER FIRST 1

- POSITIVE NUMBERS REMAIN UNCHANGED
- POSITIVE NUMBER COMPLEMENT ALWAYS START WITH 1

EXAMPLE: 8-BIT REPRESENTATION OF 12

+12: 0000 1100

-12: 1111 0100

DOES 2's COMPLEMENT HAVE TWO REPRESENTATIONS FOR 0?

2's COMPLEMENT

A BINARY REPRESENTATION OF NEGATIVE NUMBERS OBTAINED BY INVERTING ALL BITS (1's COMPLEMENT) AND ADDING 1 OR BEGINNING INVERSION AFTER FIRST 1

- POSITIVE NUMBERS REMAIN UNCHANGED
- POSITIVE NUMBER COMPLEMENT ALWAYS START WITH 1

EXAMPLE: 8-BIT REPRESENTATION OF 12

+12: 0000 1100

-12: 1111 0100

NO, IT DOES NOT, MAKING THIS NOTATION MORE WIDELY USED THAN THE OTHER TWO.

REPRESENTING REAL NUMBERS

- CONSIDER WEIGHTS AND PLACEMENT

- DECIMAL SYSTEM

- EXAMPLES: 104.32, 0.99999, 357.0, AND 3.14159

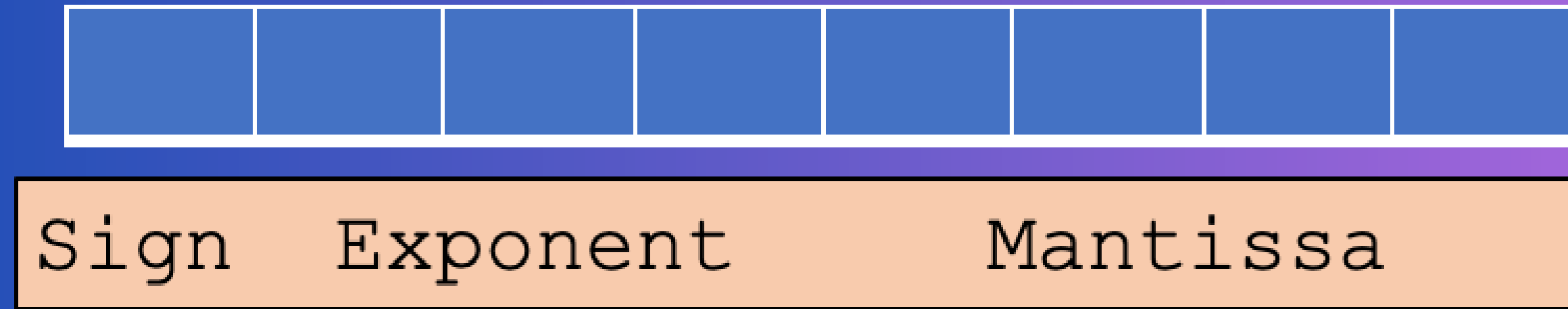
- WEIGHTS TO THE RIGHT OF THE DECIMAL ARE: $10^{-1}, 10^{-2}, 10^{-3}, \dots$

- BINARY SYSTEM

- EXAMPLES: 10.101, 0.11101, 1011.1, AND 11.111

- WEIGHTS TO THE RIGHT OF THE DECIMAL ARE: $2^{-1}, 2^{-2}, 2^{-3}, \dots$

STORING REAL NUMBERS



- **SIGN:** 0 TO INDICATE POSITIVE, 1 TO INDICATE NEGATIVE
- **MANTISSA:** THE BITS THAT RESULT FROM NORMALIZING THE NUMBER (THE RADIX POINT IS MOVED TO THE LEFT OF THE LEFTMOST 1 IN THE NUMBER)
- **EXPONENT:** INDICATES THE NUMBER OF POSITIONS THE RADIX POINT NEEDS TO BE MOVED TO THE RIGHT (IF THE EXPONENT IS POSITIVE) OR TO THE LEFT (IF THE EXPONENT IS NEGATIVE) TO GO BACK TO ITS ORIGINAL LOCATION

STORING REAL NUMBERS

EXAMPLE: $(2.625)_{10} = (10.101)_2$

NORMALIZED MANTISSA = .10101 ← THIS DIGIT IS LOST

EXPONENT = 010 (+2 IN BASE 10)



Sign	Exponent	Mantissa
------	----------	----------

WHAT IF THERE ISN'T ENOUGH ROOM FOR SOME PART OF THE REPRESENTATION (NOTABLY THE MANTISSA)?

TRUNCATION ERROR, AKA FLOATING-POINT ERROR.

CONVERTING BINARY TO DECIMAL FRACTIONS

- MULTIPLY EACH FRACTIONAL BIT BY 2 (BASE) RAISED TO THE CORRESPONDING POWER AND ADD THE PRODUCTS TOGETHER

EXAMPLE:

$$(0.101)_2 = (\quad)_{10}$$

$$1 * 2^{-1} + 0 * 2^{-2} + 1 * 2^{-3}$$

$$1 * \frac{1}{2} + 0 * \frac{1}{4} + 1 * \frac{1}{8} = 0.625$$

$$(0.101)_2 = (0.625)_{10}$$

$$2^{-1} = 0.5 = 1/2$$

REAL NUMBERS IN PROGRAMMING

VALUE	STORED AS
FLOAT	SIGN BIT, 8-BIT EXPONENT, 23-BIT MANTISSA
DOUBLE	SIGN BIT, 11-BIT EXPONENT, 52-BIT MANTISSA

ACTIVITY 5

SIGNED NUMBER SORT

- I HAVE CHOSEN SIGNED-MAGNITUDE BINARY NUMBERS
- YOU WILL CONVERT THEM TO DECIMAL NUMBERS AND THEN SORT THEM FROM SMALLEST TO LARGEST

SM BINARY NUMBERS

- 10010
- 00101
- 10100
- 00011
- 01010

ACTIVITY 5 - HELP

SIGNED NUMBER SORT

EXAMPLE : 10110

SIGN BIT: 1 -> NEGATIVE

MAGNITUDE: 0110

CONVERT MAGNITUDE TO DECIMAL:

$$(0 \cdot 2^3) + (1 \cdot 2^2) + (1 \cdot 2^1) + (0 \cdot 2^0)$$

APPLY THE NEGATIVE SIGN: -6

10110 IN SM = -6 IN DECIMAL

SM BINARY NUMBERS

- 10010
- 00101
- 10100
- 00011
- 01010

TEXT REPRESENTATION

ASCII & UNICODE

ASCII

- A CHARACTER ENCODING STANDARD THAT USES 7 OR 8 BITS(EXTENDED ASCII) TO REPRESENT 128 OR 256 CHARACTERS, INCLUDING ENGLISH LETTERS, DIGITS, AND SYMBOLS.
- MAINLY FOR SIMPLE TEXT FILES, PROGRAMMING (LIKE C-BASED LANGUAGES), AND LEGACY SYSTEMS THAT DON'T REQUIRE INTERNATIONAL CHARACTERS

UNICODE

- A UNIVERSAL CHARACTER ENCODING STANDARD THAT USES VARIABLE-LENGTH ENCODING (8, 16, OR 32 BITS) TO SUPPORT ALL LANGUAGES AND SYMBOLS WORLDWIDE.
- USED MORE WIDELY TODAY BECAUSE IT SUPPORTS CHARACTERS FROM ALL LANGUAGES, INCLUDING SYMBOLS, EMOJIS, AND SPECIAL CHARACTERS.
- THE STANDARD FOR MODERN COMPUTING, WEBSITES, AND APPLICATIONS

ASCII APPLICATIONS

- USED FOR NUMERALS
 - SOCIAL SECURITY NUMBERS
 - PHONE NUMBERS
- PLAIN, UNFORMATTED TEXT (ASCII TEXT) STORED IN TEXT FILE (.TXT)
 - APPLE: “PLAIN TEXT”
 - WINDOWS: “TEXT DOCUMENT”
- NETWORK PROTOCOLS
 - HTTP, FTP, AND EMAIL PROTOCOLS (SMTP, POP3)
- COMMAND LINE INTERFACES
- OLD SYSTEMS & DEVICES

UNICODE APPLICATIONS

- **MULTILINGUAL WEBSITES & APPS**
 - SUPPORTS LANGUAGES LIKE CHINESE, ARABIC, AND HINDI
- **EMOJIS & SYMBOLS**
 - USED IN MESSAGING APPS, SOCIAL MEDIA, AND WEB CONTENT
- **INTERNATIONAL DATABASES**
 - STORES AND PROCESSES DATA IN MULTIPLE LANGUAGES
- **MODERN OS**
 - WINDOWS, MACOS, LINUX, AND MOBILE OS USE UNICODE FOR TEXT RENDERING.
- **FONTS & TYPOGRAPHY**
 - ENABLES DIVERSE CHARACTER SETS IN DIGITAL AND PRINTED MEDIA.

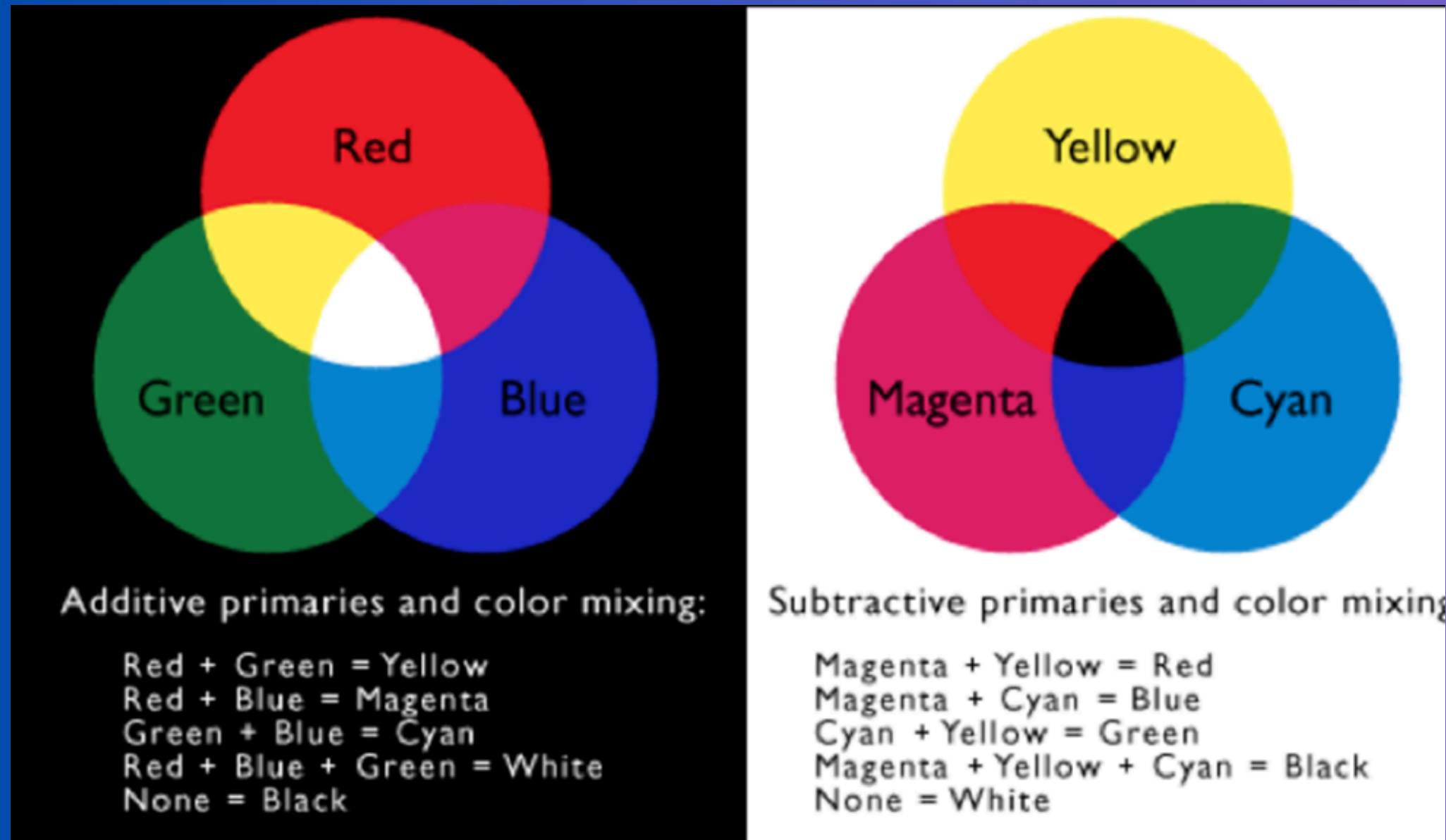
ACTIVITY 6

LOST IN TRANSLATION

- YOU WILL RESEARCH WHY ASCII ISN'T SUFFICIENT FOR GLOBAL COMMUNICATION AND HOW UNICODE SOLVED THIS PROBLEM.
- TYPE YOUR FINDINGS IN A WORD DOCUMENT.
- INCLUDE YOUR REFERENCES.
- I WILL OPEN A SUBMISSION PORTAL.

IMAGE AND GRAPHIC REPRESENTATION

COLOR

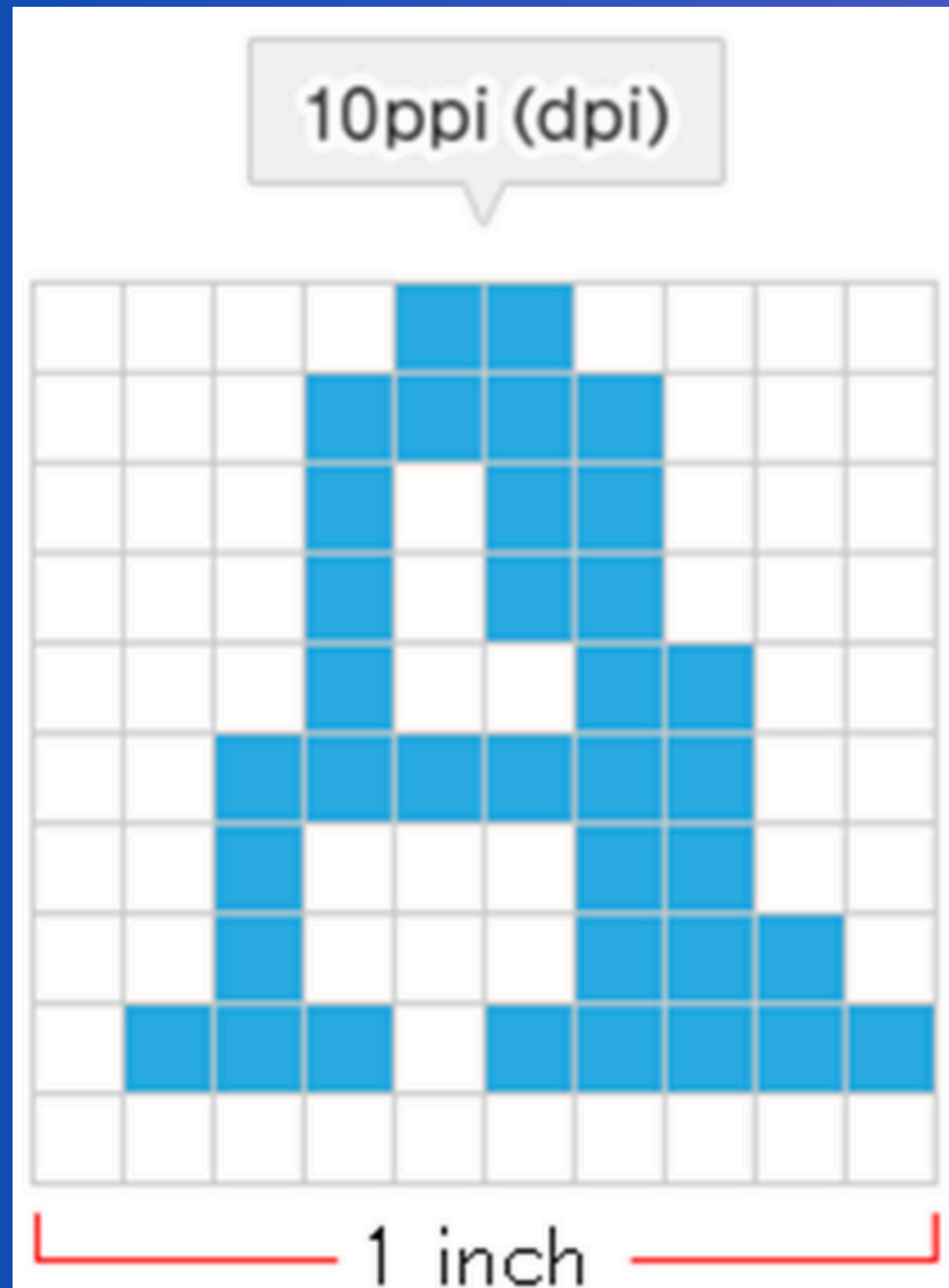


- COLOR IS EXPRESSED AS A COMBINATION OF THREE COLORS OF LIGHT
- EACH COLOR VALUE IS REPRESENTED AS ONE BYTE, SO EACH COLOR VALUE RANGES FROM 0-255
- TWO TYPES:
 - CMY (CYAN-MAGENTA-YELLOW) PRINTERS
 - RGB (RED-GREEN-BLUE) SCREEN

DIGITIZED IMAGES AND GRAPHICSS

- **DIGITIZING A PICTURE – REPRESENTING IT AS A COLLECTION OF INDIVIDUAL DOTS CALLED PIXELS**
- **RESOLUTION – THE NUMBER OF PIXELS USED TO REPRESENT A PICTURE**
- **RASTER GRAPHICS – STORAGE OF DATA ON A PIXEL-BY-PIXEL BASIS**
 - **BITMAP (BMP)- CONTAINS THE PIXEL COLOR VALUES OF THE IMAGES FROM LEFT TO RIGHT AND FROM TOP TO BOTTOM**
 - **GIF- EACH IMAGE IS MADE UP OF ONLY 256 COLORS**
 - **JPEG- AVERAGES COLOR HUES OVER SHORT DISTANCES**
 - **PNG- LIKE GIF BUT ACHIEVES GREATER COMPRESSION WITH WIDER RANGE OF COLOR DEPTHS**

STORAGE



- IMAGE SPECIFICATIONS:
 - RESOLUTION: 10PPI (10P X 10P)
 - RGB: 3 BYTES
- HOW MANY BYTES WOULD BE NEEDED TO STORE THIS IMAGE?
 - $10P \times 10P = 100$ PIXELS
 - EACH PIXEL HAS 3 BYTES 300 BYTES
 -

VECTOR GRAPHICS

- A FORMAT THAT DESCRIBES AN IMAGE IN TERMS OF LINES AND GEOMETRIC SHAPES
- A SERIES OF COMMANDS THAT DESCRIBE A LINE'S DIRECTION, THICKNESS, AND COLOR
- FILE SIZES TEND TO BE SMALLER BECAUSE NOT EVERY PIXEL IS DESCRIBED
- GOOD – RESIZED MATHEMATICALLY AND CHANGES CAN BE CALCULATED DYNAMICALLY AS NEEDED
- BAD – NOT GOOD FOR REPRESENTING REAL-WORLD IMAGES

RASTER VS VECTOR GRAPHICS

- RASTER| RESOLUTION-DEPENDENT
- VECTOR| RESOLUTION-INDEPENDENT

